

# Amritpal Singh Kaur

Graduate Software Engineer

ap.singhkaur@gmail.com · 07722 145 765 · London, UK · apsk-dev.vercel.app · LinkedIn · GitHub

## PROFESSIONAL SUMMARY

MSc Advanced Computer Science student at Queen Mary University of London (graduating 2026), with over three years of hands-on enterprise IT experience at Bank of America, Amadeus, and ENI. I design and build full-stack applications and cloud-native tools in Python, TypeScript, and React, and have independently shipped projects spanning cybersecurity dashboards, cryptography toolkits, and formal verification systems. Working in highly regulated financial and energy environments has given me a strong security-first instinct and attention to compliance — habits I carry into every codebase. I am actively looking for a 2026 graduate role in Software Engineering, Cloud/DevOps, or Cybersecurity where I can contribute from day one.

## EDUCATION

**MSc Advanced Computer Science · Queen Mary University of London** 2025 – 2026

*In Progress · Expected Merit · London, UK*

*Cloud Architecture · Machine Learning · Security & Authentication · Data Analytics · Distributed Systems · Formal Methods*

- Final Year Project: Built a CTL model checker in Python from scratch — 34/34 tests passing — covering Kripke structure parsing, temporal logic evaluation, and counterexample generation.

**BSc (Hons) Computer Science — 2:1 · Staffordshire University** 2022 – 2025

*Second Class Honours (Upper Division) · Staffordshire, UK*

*Full-Stack Development · Cloud Computing · AI & Machine Learning · Cybersecurity · Mobile Development · Databases · Networking · UX Design*

## WORK EXPERIENCE

**IT Support Engineer — Deployment & Break-Fix · Bank of America (via Manpower / Zoom IT)** Apr 2022 – Present

*London, UK*

- Owned desktop infrastructure across a heavily regulated financial environment, maintaining full compliance with FinTech security policies and GDPR — zero regulatory findings across three years.
- Diagnosed and resolved complex hardware and software incidents spanning multiple business units, consistently keeping mean time to resolution well below SLA targets.
- Delivered over 500 device deployments as part of large-scale rollout programmes, covering asset configuration, lifecycle management, and end-user handover.
- Enforced security patches, OS hardening, and configuration baselines in line with financial industry standards — proactively flagging vulnerabilities before they reached production.
- Collaborated with senior engineers and compliance teams to document processes and share knowledge, reducing repeat escalations and improving team self-sufficiency.

**Field Support Technician — Level 2 · Amadeus Global Travel Technology (via Manpower)** Cover Shifts

*London, UK*

- Managed secure laptop and mobile device replacements including full enterprise application setup, following strict compliance and data security protocols.
- Acted as Level 2 escalation point for complex incidents, coordinating cross-functionally to resolve issues efficiently and minimise business disruption.
- Maintained detailed records of system configurations, troubleshooting steps, and outcomes to support team knowledge-sharing and future incident response.

**IT Technician / Support · ENI Energy (via Manpower)** Cover Shifts

*London, UK*

- Led a Windows 11 migration across energy infrastructure systems, coordinating testing and rollout with zero critical failures or unplanned downtime.
- Provided frontline support for hardware, software, and network connectivity issues, sustaining 99.9% system uptime across critical operational environments.
- Authored structured troubleshooting documentation and built out the team knowledge base, cutting repeat escalations and accelerating onboarding for new technicians.

## KEY PROJECTS

**CTL Model Checker (LSVT)** — Python implementation of a Computation Tree Logic model checker supporting Kripke structures, full temporal logic evaluation, counterexample generation, and a benchmark suite. Submitted as MSc formal methods research — 34/34 tests passing.

**SOC Command Centre** — Enterprise security operations dashboard with real-time threat detection, live global attack map, incident response workflows, and built-in GDPR / SOC 2 / ISO 27001 compliance tracking — built with React and Node.js.

**CipherOS — Cryptography Toolkit** — Browser-native toolkit implementing AES-256-GCM encryption, SHA-3 / MD5 hashing, JWT decoding, and password-strength analysis using the Web Crypto API with no server-side dependencies.

**NetScan Pro** — Nmap / Nessus-style vulnerability scanner simulation featuring four scan modes, a 10,000+ entry CVE database (including Log4Shell and EternalBlue), and custom detection algorithms.

**Enterprise Analytics Dashboard** — Next.js analytics platform delivering real-time KPIs, AI-powered revenue forecasting at 94% confidence, live user-behaviour tracking, and interactive Recharts visualisations.

**SecureVault** — AWS-hosted SOC simulation dashboard with live threat feeds, an animated world threat map, and IAM activity charts — designed to demonstrate cloud-native security monitoring at scale.

## TECHNICAL SKILLS

|                   |                                                                                                                                                            |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Languages         | Python, JavaScript (ES6+), TypeScript, Java, C# (.NET), HTML5, CSS3                                                                                        |
| Frameworks & Libs | React, Next.js, Node.js, Express.js, REST APIs, Scikit-learn, NumPy, Pandas, Matplotlib                                                                    |
| Cloud & DevOps    | AWS (EC2, S3, Lambda, IAM, CloudWatch), Docker, Kubernetes, Azure AD, Microsoft Intune, Active Directory, CI/CD pipelines, Linux/Unix, Windows Server, Git |
| Databases         | PostgreSQL, MySQL, SQLite, MongoDB, Query Optimisation, ER Modelling                                                                                       |
| Security          | AES-256, JWT Authentication, IAM, CVE Analysis, Network Security, Secure Design Patterns, GDPR / FinTech Compliance                                        |
| AI & Data         | Machine Learning, NLP, Statistical Analysis, Predictive Modelling, Data Visualisation, Model Evaluation                                                    |
| Methods           | Agile / Scrum, RESTful API Design, OOP, Functional Programming, Formal Verification                                                                        |
| Languages Spoken  | English · Spanish · Catalan · Punjabi (Native) · Hindi (Professional)                                                                                      |